1) a)

$$\begin{pmatrix}2 & 1 & 0\\ 1 & 2 & 0\\ 0 & 0 & 4\end{pmatrix}\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}=\lambda\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}3\\ 3\\ 0\end{pmatrix}=\lambda\begin{pmatrix}1\\ 1\\ 0\end{pmatrix}$$

$$\therefore \lambda=3$$

1) b)

$$\begin{pmatrix}1 & -2 & 1\\ 1 & 1 & 1\\ 1 & 2 & 2\end{pmatrix}\begin{pmatrix}0\\ 1\\ 2\end{pmatrix}=\lambda\begin{pmatrix}0\\ 1\\ 2\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}0\\ 3\\ 6\end{pmatrix}=\lambda\begin{pmatrix}0\\ 1\\ 2\end{pmatrix}$$

$$\therefore \lambda=3$$

1) c)

$$\begin{pmatrix}-7 & 1 & 0\\ -16 & 0 & 2\\ -7 & -1 & 1\end{pmatrix}\begin{pmatrix}1\\ 3\\ 2\end{pmatrix}=\lambda\begin{pmatrix}1\\ 3\\ 2\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}-4\\ -12\\ -8\end{pmatrix}=\lambda\begin{pmatrix}1\\ 3\\ 2\end{pmatrix}$$

$$\therefore \lambda=-4$$

2) a)

$$\begin{pmatrix}-11 & -6 & 10\\ -4 & -9 & 8\\ -10 & -12 & 14\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}-11 & -6 & 10\\ -4 & -9 & 8\\ -10 & -12 & 14\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}-11-\lambda & -6 & 10\\ -4 & -9-\lambda & 8\\ -10 & -12 & 14-\lambda\end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I})=\begin{vmatrix}-11-\lambda & -6 & 10\\ -4 & -9-\lambda & 8\\ -10 & -12 & 14-\lambda\end{vmatrix}=0$$

$$\begin{vmatrix}-11-\lambda & -6 & -2\lambda-12\\ -4 & -9-\lambda & 0\\ -10 & -12 & -\lambda-6\end{vmatrix}=0$$

$\det(\mathbf{M}-\lambda\mathbf{I})=(-2\lambda-12)(-10\lambda-42)+(-\lambda-6)(\lambda^2+20\lambda+75)$

$=(\lambda+6)(20\lambda+84-\lambda^2-20\lambda-75)$

$=(\lambda+6)(9-\lambda^2)$

$\therefore(\lambda+6)(3-\lambda)(3+\lambda)=0\Rightarrow(\lambda-3)(\lambda+3)(\lambda+6)=0$ as req.

2) b)

$-6, -3$ and 3

3) a)

$$\begin{pmatrix}2 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & -3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}2 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & -3\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}2-\lambda & 1 & 1\\ 0 & 1-\lambda & 1\\ 0 & 0 & -3-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(2-\lambda)(1-\lambda)(-3-\lambda)-0+0$

So $(2-\lambda)(1-\lambda)(-3-\lambda)=0$

$\therefore\lambda=2,\ \lambda=1$ or $\lambda=-3$

3) b)

$$\begin{pmatrix}1 & -2 & 0\\ 2 & 1 & 5\\ 0 & 0 & 5\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1 & -2 & 0\\ 2 & 1 & 5\\ 0 & 0 & 5\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}1-\lambda & -2 & 0\\ 2 & 1-\lambda & 5\\ 0 & 0 & 5-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=0-0+(5-\lambda)((1-\lambda)^2+4)$

So $(5-\lambda)(\lambda^2-2\lambda+5)=0$

$\therefore \lambda=5,\ \lambda=1\pm 2\mathrm{i}$

3) c)

$$\begin{pmatrix}0 & -1 & 1\\ -1 & -1 & 2\\ -1 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}0 & -1 & 1\\ -1 & -1 & 2\\ -1 & 0 & 1\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}-\lambda & -1 & 1\\ -1 & -1-\lambda & 2\\ -1 & 0 & 1-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=-\lambda\big((-1-\lambda)(1-\lambda)-0\big)+(-(1-\lambda)+2)+(-1-\lambda)$

$=-\lambda(\lambda^2-1)+\lambda+1-1-\lambda$

So $-\lambda(\lambda-1)(\lambda+1)=0$

$\therefore \lambda=0,\ \lambda=1$ or $\lambda=-1$

3) d)

$$\begin{pmatrix}1 & -1 & -3\\ 0 & 4 & -8\\ 0 & 4 & -4\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1 & -1 & -3\\ 0 & 4 & -8\\ 0 & 4 & -4\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}1-\lambda & -1 & -3\\ 0 & 4-\lambda & -8\\ 0 & 4 & -4-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I})=(1-\lambda)\begin{vmatrix}4-\lambda & -8\\ 4 & -4-\lambda\end{vmatrix}-0+0$

$=(1-\lambda)\big((4-\lambda)(-4-\lambda)+32\big)$

$=(1-\lambda)(\lambda^2-16+32)$

So $(1-\lambda)(\lambda^2+16)=0$

$\therefore \lambda=1$ or $\lambda=\pm 4\mathrm{i}$

4)

$$\begin{pmatrix} p & -q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & -q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & -q & 0 \\ q & p-\lambda & 0 \\ 0 & 0 & r-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} p-\lambda & -q & 0 \\ q & p-\lambda & 0 \\ 0 & 0 & r-\lambda \end{vmatrix} = 0$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (r-\lambda)\big((p-\lambda)(p-\lambda) + q^2\big)$$

$$= (r-\lambda)(\lambda^2 - 2p\lambda + p^2 + q^2)$$

$$\lambda^2 - 2p\lambda + p^2 + q^2 = 0$$

$$\Rightarrow \lambda = \frac{2p \pm \sqrt{(-2p)^2 - 4(p^2+q^2)}}{2} = \frac{2p \pm \sqrt{4p^2 - 4p^2 - 4q^2}}{2} = \frac{2p \pm \sqrt{-4q^2}}{2} = \frac{2p \pm 2\sqrt{-q^2}}{2} = p \pm q\text{i}$$

$\therefore \lambda = r$ and $\lambda = p \pm q\text{i}$

5) a)

$$\begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-\lambda & 0 & -\frac{3}{2} \\ 0 & -5-\lambda & \frac{11}{2} \\ 0 & 0 & \frac{1}{2}-\lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(-5-\lambda)\left(\frac{1}{2}-\lambda\right)$$

$\therefore \lambda = 2$, $\lambda = -5$ and $\lambda = \frac{1}{2}$

5) b)

$$\mathbf{M}^2 = \begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$$

$$\begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4-\lambda & 0 & -\frac{15}{4} \\ 0 & 25-\lambda & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4}-\lambda \end{pmatrix}$$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(25-\lambda)\left(\frac{1}{4}-\lambda\right)$

$\therefore \lambda = 4$, $\lambda = 25$ and $\lambda = \frac{1}{4}$

5) c)

The eigenvalues of $\mathbf{A}$ are $\lambda = 2$, $\lambda = -5$ and $\lambda = \frac{1}{2}$

$\Rightarrow \lambda^2 = 4$, $\lambda^2 = 25$ and $\lambda^2 = \frac{1}{4}$ which are the eigenvalues of $\mathbf{A}^2$

5) d)

If $\mathbf{Mv} = \lambda\mathbf{v}$, then

$\mathbf{M}^2\mathbf{v} = \mathbf{M}(\mathbf{Mv}) = \mathbf{M}(\lambda\mathbf{v}) = \lambda\mathbf{Mv} = \lambda(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$ as req.

6) a)

$$\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}3-\lambda & -3 & 1\\ 0 & -5-\lambda & 6\\ 0 & -3 & 4-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}3-\lambda & -3 & 1\\ 0 & -5-\lambda & 6\\ 0 & -3 & 4-\lambda\end{vmatrix}=0$$

$$=(3-\lambda)\big((-5-\lambda)(4-\lambda)+18\big)$$

$$=(3-\lambda)(\lambda^2+\lambda-2)$$

$$=(3-\lambda)(\lambda+2)(\lambda-1)$$

$\therefore \lambda=-2$, $\lambda=1$ and $\lambda=3$

6) b)

$$\mathbf{M}^3=\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}=\begin{pmatrix}3 & -3 & 1\\ 0 & -5 & 6\\ 0 & -3 & 4\end{pmatrix}\begin{pmatrix}9 & 3 & -11\\ 0 & 7 & -6\\ 0 & 3 & -2\end{pmatrix}$$

$$=\begin{pmatrix}27 & -9 & -17\\ 0 & -17 & 18\\ 0 & -9 & 10\end{pmatrix}$$

$$\begin{pmatrix}27 & -9 & -17\\ 0 & -17 & 18\\ 0 & -9 & 10\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}27 & -9 & -17\\ 0 & -17 & 18\\ 0 & -9 & 10\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}27-\lambda & -9 & -17\\ 0 & -17-\lambda & 18\\ 0 & -9 & 10-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}27-\lambda & -9 & -17\\ 0 & -17-\lambda & 18\\ 0 & -9 & 10-\lambda\end{vmatrix}=0$$

$$=(27-\lambda)\big((-17-\lambda)(10-\lambda)+162\big)$$

$$=(27-\lambda)(\lambda^2+7\lambda-8)$$

$$=(27-\lambda)(\lambda+8)(\lambda-1)$$

$\therefore \lambda=-8$, $\lambda=1$ and $\lambda=27$

6) c)

The eigenvalues of $\mathbf{A}$ are $\lambda=-2$, $\lambda=1$ and $\lambda=3$

$\Rightarrow \lambda^3=-8$, $\lambda^3=1$ and $\lambda^3=27$ which are the eigenvalues of $\mathbf{A}^3$

6) d)

If $\mathbf{Mv} = \lambda\mathbf{v}$, then

$\mathbf{M}^2\mathbf{v} = \mathbf{M}(\mathbf{Mv}) = \mathbf{M}(\lambda\mathbf{v}) = \lambda\mathbf{Mv} = \lambda(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$

$\Rightarrow \mathbf{M}^3\mathbf{v} = \mathbf{M}(\mathbf{M}^2\mathbf{v}) = \mathbf{M}(\lambda^2\mathbf{v}) = \lambda^2(\mathbf{Mv}) = \lambda^2(\lambda\mathbf{v}) = \lambda^3\mathbf{v}$ as req.

7) a)

$$\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} + 3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -23 & 8 & 8 \\ -72 & 25 & 24 \\ -24 & 8 & 9 \end{pmatrix} + 3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -20 & 8 & 8 \\ -72 & 28 & 24 \\ -24 & 8 & 12 \end{pmatrix}$$

$$k\begin{pmatrix} -20 & 8 & 8 \\ -72 & 28 & 24 \\ -24 & 8 & 12 \end{pmatrix} = 4\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}$$

$\therefore k = 4$

7) b)

$\mathbf{M}^2 + 3\mathbf{I} = 4\mathbf{M}$

$\mathbf{M}^2 - 4\mathbf{M} + 3\mathbf{I} = \mathbf{0}$

$\mathbf{M} - 4\mathbf{I} + 3\mathbf{M}^{-1} = \mathbf{0}$

$3\mathbf{M}^{-1} = 4\mathbf{I} - \mathbf{M}$

$\mathbf{M}^{-1} = \frac{4}{3}\mathbf{I} - \frac{1}{3}\mathbf{M}$

$= -\frac{1}{3}\mathbf{M} + \frac{4}{3}\mathbf{I}$ as req.

7) c)

$$\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -5-\lambda & 2 & 2 \\ -18 & 7-\lambda & 6 \\ -6 & 2 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} -5-\lambda & 2 & 2 \\ -18 & 7-\lambda & 6 \\ -6 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -5-\lambda & 0 & 2 \\ -18 & 1-\lambda & 6 \\ -6 & -1+\lambda & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -5-\lambda & 0 & 2 \\ -24 & 0 & 9-\lambda \\ -6 & -1+\lambda & 3-\lambda \end{vmatrix} = 0$$

$$= -(-1+\lambda)\big((-5-\lambda)(9-\lambda) - (2)(-24)\big)$$

$$= -(-1+\lambda)(\lambda^2 - 4\lambda - 45 + 48)$$

$$= -(-1+\lambda)(\lambda-1)(\lambda-3)$$

$\therefore \lambda = 1$, $\lambda = 1$ and $\lambda = 3$

7) d)

$1^2 + 3 = 4 = 4 \times 1$ LHS=RHS $\therefore$ true

$3^2 + 3 = 12 = 4 \times 3$ LHS=RHS $\therefore$ true

8) a)

$$\begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 3 & -1 \\ -2 & 3-\lambda & 2 \\ 1 & 0 & 2-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 3 & -1 \\ -2 & 3-\lambda & 2 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 3 & -\lambda \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 3 & -3 \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 3 & -3 \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = -3(\lambda - 3) - 0 + (3-\lambda)(\lambda^2 - 3\lambda + 6)$$

$$= (3-\lambda)(3 + \lambda^2 - 3\lambda + 6)$$

$\therefore (\lambda - 3)(\lambda^2 - 3\lambda + 9) \ = 0$ as req.

8) b)

$\lambda = 3$ and $\lambda = \frac{3}{2} \pm \frac{3\sqrt{3}}{2}\mathrm{i}$

When $\lambda = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} -2 & 3 & -1 \\ -2 & 0 & 2 \\ 1 & 0 & -1 \end{pmatrix}$

$$\begin{pmatrix} -2 & 3 & -1 \\ -2 & 0 & 2 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -2x + 3y - z = 0$ and $-2x + 2z = 0$ and $-x - z = 0$

So $x = z$ and so $y = x = z$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

9) a)

When $\lambda = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 4 & 3 & 6 \\ -2 & -5 & -10 \\ -8 & -3 & -6 \end{pmatrix}$

$$\begin{pmatrix} 4 & 3 & 6 \\ -2 & -5 & -10 \\ -8 & -3 & -6 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 4x + 3y + 6z = 0$ (1)

$-2x - 5y - 10z = 0$ (2)

$-8x - 3y - 6z = 0$ (3)

(1) $+2 \times$(2): $-7y - 14z = 0 \ \therefore y = -2z$

Sub into (1): $4x = 0 \therefore x = 0$

$\therefore$ when $\lambda = 2$, $\mathbf{v} = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$

9) b)

When $\lambda = 0$ then $\mathbf{A} - 0\mathbf{I} = \begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$

$$\begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -2x + 5y + 7z = 0$ (1)

$x - z = 0$ (2)

$-x + y + 2z = 0$ (3)

From (2): $x = z$

Sub into (3): $y + x = 0 \therefore -y = x$

$\therefore$ when $\lambda = 0$, $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

9) c)

When $\lambda = 1$ then $\mathbf{A} + 1\mathbf{I} = \begin{pmatrix} 5 & -8 & 3 \\ 3 & -6 & 3 \\ 4 & -10 & 6 \end{pmatrix}$

$$\begin{pmatrix} 5 & -8 & 3 \\ 3 & -6 & 3 \\ 4 & -10 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 5x - 8y + 3z = 0$ (1)

$3x - 6y + 3z = 0$ (2)

$4x - 10y + 6z = 0$ (3)

(2) − (1): $2y - 2x = 0 \therefore y = x$

Sub into (1): $-3x + 3z = 0 \therefore z = x$

$\therefore$ when $\lambda = 1$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

9) d)

When $\lambda = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 2 \\ 4 & 2 & -1 \\ -4 & 1 & 3 \end{pmatrix}$

$$\begin{pmatrix} -4 & 1 & 2 \\ 4 & -1 & -1 \\ -4 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -4x + y + 2z = 0$ (1)

$4x - y - z = 0$ (2)

$-4x + y = 0$ (3)

From (3): $y = 4x$

Sub into (1): $2z = 0 \therefore z = 0$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}$

9) e)

When $\lambda = 3$ then $\mathbf{A} + 5\mathbf{I} = \begin{pmatrix} 0 & 1 & 1 \\ \frac{3}{2} & 5 & 2 \\ -1 & 3 & 5 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 & 1 \\ \frac{3}{2} & 5 & 2 \\ -1 & 3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow y + z = 0$ (1)

$\frac{3}{2}x + 5y + 2z = 0$ (2)

$-x + 3y + 5z = 0$ (3)

From (1): $y = -z$

Sub into (3): $-x + 2z = 0 \therefore 2z = x$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

10) a)

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 & 0 \\ 1 & 4-\lambda & 0 \\ 1 & 1 & 5-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = 0 - 0 - (5-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix}$$

$= (\lambda - 5)(\lambda^2 - 8\lambda + 15)$

So $(\lambda - 5)(\lambda - 5)(\lambda - 3) = 0$

So $\lambda = 5$ is a repeated eigenvalue of $\mathbf{M}$

10) b)

$\lambda_1 = 5$, $\lambda_2 = 5$ or $\lambda_3 = 3$

10) c)

When $\lambda_1 = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$

$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow -x + y = 0$ and $x + y = 0$

So $x = 0$ and $y = 0$ and z may take any value

$\therefore$ when $\lambda_1 = 5$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

When $\lambda_3 = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow x + y = 0$ and $x + y + 2z = 0$

So $y = -x$ and $z = 0$ and z may take any value

$\therefore$ when $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$\therefore \lambda_1 = 5$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ and $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

11) a)

$$\begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 6-\lambda & 1 & -4 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} 6-\lambda & 1 & -4 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & -3+\lambda & 0 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & 0 & 0 \\ 3 & 7-\lambda & -4 \\ 7 & 8 & -5-\lambda \end{vmatrix} = 0$$

$$= (3-\lambda)\begin{vmatrix} 7-\lambda & -4 \\ 8 & -5-\lambda \end{vmatrix}$$

$$= (3-\lambda)(\lambda^2 - 2\lambda - 3)$$

So $\lambda = 3$ is an eigenvalue of $\mathbf{M}$

11) b)

$(\lambda - 3)(\lambda^2 - 2\lambda - 3) = (\lambda - 3)(\lambda - 3)(\lambda + 1)$

$\lambda_1 = 3$, $\lambda_2 = 3$ or $\lambda_3 = -1$

11) c)

When $\lambda_1 = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} 3 & 1 & -4 \\ 3 & 1 & -4 \\ 7 & 1 & -8 \end{pmatrix}$

$$\begin{pmatrix} 3 & 1 & -4 \\ 3 & 1 & -4 \\ 7 & 1 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 3x + y - 4z = 0$ (1)

$7x + y - 8z = 0$ (2)

(2) − (1): $4x - 4z = 0 \therefore x = z$

Sub into (1): $-x + y = 0 \therefore x = y$

$\therefore$ when $\lambda_1 = 3$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

When $\lambda_3 = -1$ then $\mathbf{A} + 1\mathbf{I} = \begin{pmatrix} 7 & 1 & -4 \\ 3 & 5 & -4 \\ 7 & 1 & -4 \end{pmatrix}$

$$\begin{pmatrix} 7 & 1 & -4 \\ 3 & 5 & -4 \\ 7 & 1 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 7x + y - 4z = 0$ (1)

$3x + 5y - 4z = 0$ (2)

(2) − (1): $-4x + 4y = 0 \therefore y = x$

Sub into (1): $8x - 4z = 0 \therefore z = 2x$

$\therefore$ when $\lambda_3 = -1$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

$\therefore$ when $\lambda_1 = 3$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\lambda_3 = -1$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

12) a)

$$\begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -4-\lambda & 16 & -8 \\ -5 & 13-\lambda & -6 \\ -10 & 22 & -10-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} -4-\lambda & 16 & -8 \\ -5 & 13-\lambda & -6 \\ -10 & 22 & -10-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -4-\lambda & 0 & -8 \\ -5 & 1-\lambda & -6 \\ -10 & 2-2\lambda & -10-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -4-\lambda & 0 & -8 \\ -5 & 1-\lambda & -6 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$= (2-\lambda)(-4-\lambda)(1-\lambda)$

So $\lambda = 2$ is an eigenvalue of $\mathbf{M}$

12) b)

$(2-\lambda)(-4-\lambda)(1-\lambda) = 0$

$\lambda_1 = 2$, $\lambda_2 = -4$ or $\lambda_3 = 1$

12) c)

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -6 & 16 & -8 \\ -5 & 11 & -6 \\ -10 & 22 & -12 \end{pmatrix}$

$$\begin{pmatrix} -6 & 16 & -8 \\ -5 & 11 & -6 \\ -10 & 22 & -12 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -6x + 16y - 8z = 0$ (1)

$-5x + 11y - 6z = 0$ (2)

$5 \times$ (1) $\Rightarrow -30x + 80y - 40z = 0$ (3)

$6 \times$ (2) $\Rightarrow -30x + 66y - 36z = 0$ (4)

(3) $-$ (4): $14y - 4z = 0 \therefore 7y = 2z$

Sub into (1): $-6x + 16y - 28y = 0 \Rightarrow -6x - 12y = 0 \therefore -2y = x$

$\therefore$ when $\lambda_1 = 2$, $\mathbf{v}_1 = \begin{pmatrix} 4 \\ -2 \\ -7 \end{pmatrix}$

When $\lambda_2 = -4$ then $\mathbf{A} + 4\mathbf{I} = \begin{pmatrix} 0 & 16 & -8 \\ -5 & 17 & -6 \\ -10 & 22 & -6 \end{pmatrix}$

$\begin{pmatrix} 0 & 16 & -8 \\ -5 & 17 & -6 \\ -10 & 22 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow 16y - 8z = 0$ (1)

$-5x + 17y - 6z = 0$ (2)

$-10x + 22y - 6z = 0$ (3)

From (1): $2y = z$

Sub into (2): $-5x + 17y - 12y = 0 \Rightarrow -5x + 5y = 0 \therefore y = x$

$\therefore$ when $\lambda_2 = -4$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

When $\lambda_3 = 1$ then $\mathbf{A} - 1\mathbf{I} = \begin{pmatrix} -5 & 16 & -8 \\ -5 & 12 & -6 \\ -10 & 22 & -11 \end{pmatrix}$

$\begin{pmatrix} -5 & 16 & -8 \\ -5 & 12 & -6 \\ -10 & 22 & -11 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow -5x + 16y - 8z = 0$ (1)

$-5x + 12y - 6z = 0$ (2)

(1) $-$ (2): $4y - 2z = 0 \therefore z = 2y$

Sub into (1): $-5x = 0 \therefore x = 0$

$\therefore$ when $\lambda_3 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

$\therefore$ when $\lambda_1 = 2$, $\mathbf{v}_1 = \begin{pmatrix} 4 \\ -2 \\ -7 \end{pmatrix}$, $\lambda_2 = -4$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, $\lambda_3 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

13)

Since $\mathbf{Mv} = \lambda\mathbf{v}$

$$\mathbf{Mv} = \begin{pmatrix} -1 & p & 0 \\ p & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix} = \begin{pmatrix} -1+2p \\ p+4 \\ 4+q \end{pmatrix}$$

$$\lambda\mathbf{v} = 3\begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 3q \end{pmatrix}$$

$-1+2p = 3 \therefore p = 2$

$4+q = 3q \therefore q = 2$

14) a)

$$\frac{1}{14}\begin{pmatrix} 10 & -44 & -2k \\ 0 & 42 & 0 \\ 6 & -53 & -17 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 5-2k \\ 0 \\ -14 \end{pmatrix}$$

$$\Rightarrow \lambda_3\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 5-2k \\ 0 \\ -14 \end{pmatrix}$$

$2\lambda_3 = -2 \therefore \lambda_3 = -1$

$-1 = \frac{5-2k}{7} \Rightarrow -12 = -2k \Rightarrow \ \therefore k = 6$

14) b)

$$\frac{1}{14}\begin{pmatrix} 10 & -44 & -12 \\ 0 & 42 & 0 \\ 6 & -53 & -17 \end{pmatrix}\begin{pmatrix} a \\ 0 \\ 1 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 10a-12 \\ 0 \\ 6a-17 \end{pmatrix}$$

$$\Rightarrow \lambda_2\begin{pmatrix} a \\ 0 \\ 1 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 10a-12 \\ 0 \\ 6a-17 \end{pmatrix}$$

$14a\lambda_2 = 10a-12$ and $\lambda_2 = \frac{6a-17}{14}$

So that $14a\left(\frac{6a-17}{14}\right) = 10a-12 \Rightarrow 6a^2-17a = 10a-12 \Rightarrow 6a^2-27a+12 = 0$

$a = \frac{1}{2}$ or $a = 4$ but a is an integer.

$\therefore a = 4$

14) c)

$\lambda_2 = \frac{6a-17}{14} = \frac{24-17}{14} = \frac{1}{2}$

$\therefore \lambda_1 = 3,\ \lambda_2 = \frac{1}{2}, \lambda_3 = -1$

15) a)

$$\begin{pmatrix} 6 & -2 & 2 \\ 2-k & 2+k & k-2 \\ 4-k & k-4 & k+4 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} 6 & -2 & 2 \\ 2-k & 2+k & k-2 \\ 4-k & k-4 & k+4 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 6-\lambda & -2 & 2 \\ 2-k & 2+k-\lambda & k-2 \\ 4-k & k-4 & k+4-\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} 6-\lambda & -2 & 2 \\ 2-k & 2+k-\lambda & k-2 \\ 4-k & k-4 & k+4-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & 0 \\ 2-k & 2+k-\lambda & 2k-\lambda \\ 4-k & k-4 & 2k-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 6-\lambda & 0 \\ 4-k & k-4 & 2k-\lambda \end{vmatrix} = 0$$

$$= (2k-\lambda)\begin{vmatrix} 6-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix}$$

$$= (2k-\lambda)(\lambda^2 - 12\lambda + 32) = (2k-\lambda)(\lambda-4)(\lambda-8)$$

So $\lambda = 2k$ is an eigenvalue of $\mathbf{M}$

15) b)

When $\lambda_3 = 2k$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} 6-2k & -2 & 2 \\ 2-k & 2-k & k-2 \\ 4-k & k-4 & 4-k \end{pmatrix}$

$\begin{pmatrix} 6-2k & -2 & 2 \\ 2-k & 2-k & k-2 \\ 4-k & k-4 & 4-k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$\Rightarrow (6-2k)x - 2y + 2z = 0$ (1)

$(2-k)x + (2-k)y + (k-2)z = 0$ (2)

$(4-k)x + (k-4)y + (4-k)z = 0$ (3)

From (2): $(2-k)(x+y-z) = 0 \Rightarrow x + y - z = 0$

From (3): $(4-k)(x-y+z) = 0 \Rightarrow x - y + z = 0$

$\therefore x = 0$ and $y = z$

$\therefore$ when $\lambda_3 = 2k$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

16) a)

$\begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} k-2-\lambda & -3 & 3 \\ 6 & -5-\lambda & 3 \\ 0 & 0 & -2-\lambda \end{pmatrix}$

$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} k-2-\lambda & -3 & 3 \\ 6 & -5-\lambda & 3 \\ 0 & 0 & -2-\lambda \end{vmatrix} = 0$

$= (-2-\lambda)\begin{vmatrix} k-2-\lambda & -3 \\ 6 & -5-\lambda \end{vmatrix}$

$= -(\lambda+2)\big((k-2-\lambda)(-5-\lambda) - (-18)\big)$

$= -(\lambda+2)(\lambda^2 + (7-k)\lambda + 28 - 5k) = 0$

So consider the quadratic: $b^2 - 4ac = 0$ creates a repeated root

$(7-k)^2 - 4(28-5k) = 0 \Rightarrow k^2 - 14k + 49 + 20k - 112 = 0$

$\Rightarrow k^2 + 6k - 63 = 0$ so $k = -3 \pm 6\sqrt{2}$ which is not an integer.

So $\lambda = -2$ must be the repeated root.

When $\lambda = -2$ then

$4 - 2(7-k) + 28 - 5k = 0 \Rightarrow 4 - 14 + 2k + 28 - 5k = 0$

$\therefore k = 6$

16) b)

When $k = 6$

$-(\lambda + 2)(\lambda^2 + \lambda - 2) = -(\lambda + 2)(\lambda + 2)(\lambda - 1)$

So eigenvalues are $\lambda_1 = -2$, $\lambda_2 = -2$ and $\lambda_3 = 1$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 6 & -3 & 3 \\ 6 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 6 & -3 & 3 \\ 6 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 6x - 3y + 3z = 0 \quad (1)$

So $x = 0$ gives $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $y = 0$ gives $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ and $z = 0$ gives $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

Any two linearly independent ones span the eigenspace for $\lambda = -2$

Since $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ then $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ is dependent. So choose $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -2$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

When $\lambda_2 = 1$ then $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 3 & -3 & 3 \\ 6 & -6 & 3 \\ 0 & 0 & -3 \end{pmatrix}$

$$\begin{pmatrix} 3 & -3 & 3 \\ 6 & -6 & 3 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow 3x - 3y + 3z = 0 \quad (1)$

$6x - 6y + 3z = 0 \quad (2)$

(2) − (1): $3x - 3y = 0 \qquad \therefore x = y$

From (1): $3z = 0 \qquad \therefore z = 0$

$\therefore$ when $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -2$, $\mathbf{v}_1 = \text{span}\left\{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right\}$, $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

16) c)

$\text{trace}(\mathbf{M}) = 4 - 5 - 2 = -3$

17) a)

$$\det \mathbf{M} = \begin{vmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{vmatrix}$$

$$= \begin{vmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ -c & 0 & c \end{vmatrix}$$

$$= \begin{vmatrix} -c+2a & -a+b & -a+c \\ 0 & b & 0 \\ -c & 0 & c \end{vmatrix}$$

$$= \begin{vmatrix} a & -a+b & -a+c \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix}$$

$$= a\begin{vmatrix} b & 0 \\ 0 & c \end{vmatrix} - 0 + 0$$

$$= abc$$

17) b)

$$\begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M}-\lambda\mathbf{I} = \begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ 3a-b-2c & -a+b & -a+2c-\lambda \end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I}) = \begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ 3a-b-2c & -a+b & -a+2c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ \lambda-c & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ \lambda-b & b-\lambda & 0 \\ \lambda-c & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-b-\lambda & -a+b & -a+c \\ \lambda-b & b-\lambda & 0 \\ 0 & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a-\lambda & -a+b & -a+c \\ 0 & b-\lambda & 0 \\ 0 & 0 & c-\lambda \end{vmatrix} = 0$$

$$= (a-\lambda)\begin{vmatrix} b-\lambda & 0 \\ 0 & c-\lambda \end{vmatrix} - 0 + 0$$

$$= (a-\lambda)(b-\lambda)(c-\lambda)$$

So the eigenvalues are $\lambda = a$, $\lambda = b$ and $\lambda = c$

17) c)

$$3a-b-c-\lambda-a+2b-\lambda+c-\lambda = a+b+c$$

18) a)

$$\det\left(\frac{1}{2}\mathbf{M}\right) = \left(\frac{1}{2}\right)^3 \det(\mathbf{M})$$

$$\frac{1}{8}\begin{vmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} a+c & a-c & 2c \\ a-b & a+b & 0 \\ c-b & b-c & 2c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} a+b & a-b & 0 \\ a-b & a+b & 0 \\ c-b & b-c & 2c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} 2b & 0 & 0 \\ a-b & 2a & 0 \\ c-b & 0 & 2c \end{vmatrix}$$

$$= \frac{1}{8}(2b)(2a)(2c) = abc$$

18) b)

$$\frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} a+c-2\lambda & a-c & c-a \\ a-b & a+b-2\lambda & b-a \\ c-b & b-c & b+c-2\lambda \end{pmatrix}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \frac{1}{2}\begin{vmatrix} a+c-2\lambda & a-c & c-a \\ a-b & a+b-2\lambda & b-a \\ c-b & b-c & b+c-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a+c-2\lambda & a-c & 0 \\ a-b & a+b-2\lambda & 2b-2\lambda \\ c-b & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a+c-2\lambda & a-c & 0 \\ a-c & a+c-2\lambda & 0 \\ c-b & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-2\lambda & a-c & 0 \\ 2a-2\lambda & a+c-2\lambda & 0 \\ 0 & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-2\lambda & a-c & 0 \\ 0 & 2c-2\lambda & 0 \\ 0 & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$= (2a-2\lambda)(2c-2\lambda)(2b-2\lambda)$$

So the eigenvalues are $\lambda = a$, $\lambda = b$ and $\lambda = c$

18) c)

$$\text{trace}(\mathbf{M}) = \frac{1}{2}(a+c+a+b+b+c) = a+b+c$$